

K3 ENCRYPTION MATRIX

While it may or may not be the actual matrix Jim Sanborn used, the simplest method I found for encrypting K3 was to pad the plaintext with a seven-letter keyword (i.e., kryptos) and evenly distribute the text across an 86x4 grid. The ciphertext is readily obtained in reading every seventh column, upward, from right to left.

Encryption Matrix:

K	R	Y	P	T	O	S	S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S	O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D
O	O	R	W	A	Y	W	A	S	R	E	M	O	V	E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N	D	T	H	E	N	W	I	D	E	N
I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I	N	T	H	E	H	O	T	A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	T	H	E
F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T	A	I	L	S	O	F	T	H	E	R	O	O	M	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	T	H	E	M	I	S	T	X	C	A	N	Y	O	U	S	E	E	A	N	Y	T	H	I	N	G	Q	?

Ciphertext:

?END YAHR OHNL SRHE OCPT EOIB IDYS HNAI ACHT NREY ULDS LLSL LNOR
HSNO SMRW XMNE TPRN GATI HNRA RPES LNNE LEBL PIIA CAEW MTWP
NDIT EENR AHCT ENEU DRET NHA E OETF OLSE DTIW ENHA EIOY TEYO
QHEE NCTA YCRE IFTB RSPA MHHE WENA TAMA TEGY EERL BTEE FOAS FIOK
TUET UAO TOAR MAEE RTNR TIBS EDDN IAAH TTMS TEWP IERO AGRY
IEWF EB AE CTDD HILC EIHS ITEG OEAO SDDR YDLO RITR KLML EHAT
GTDH ARDP NEOH MGFM FEUH EEC D MRIP FEIM EHNL SST RTVD OHWS

You'll notice, however, there's a discrepancy between the ciphertext generated by this matrix and that found on the sculpture. Specifically, the position of the question mark. I suppose Jim could've transposed it post-encryption, but why? If it were meant to make the cipher more difficult to crack, a character deeper in the text would've been a better choice. (see *Alternative Transposition Test*) Further, if "Q" is shorthand for "Question", as I understood it, the question mark is redundant. Why end a sentence with double punctuation?

With a slight modification to the matrix above, the answer presents itself.

Revised Encryption Matrix:

K	R	Y	P	T	O	S	?S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S	O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	
D	O	O	R	W	A	Y	W	A	S	R	E	M	O	V	E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N	D	T	H	E	N	W	I	D	E	N
I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I	N	T	H	E	H	O	T	A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	T	H	E	
E	F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T	A	I	L	S	O	F	T	H	E	R	O	O	M	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	T	H	E	M	I	S	T	X	C	A	N	Y	O	U	S	E	E	A	N	Y	T	H	I	N	G	Q	?

Ciphertext:

ENDK YAHR OHNL SRHE OCPT EOIB IDYS HNAI ACHT NREY ULDS LLSL LNOY
HSNO SMRW XMNE TPRN GATI HNRA RPES LNNE LEBL PIIA CAEW MTWT
NDIT EENR AHCT ENEU DRET NHA E OETF OLSE DTIW ENHA EIOY TEYS
QHEE NCTA YCRE IFTB RSPA MHHE WENA TAMA TEGY EERL BTEE FOAS FIOR
TUET UAO TOAR MAEE RTNR TIBS EDDN IAAH TTMS TEWP IERO AGRP
IEWF EB AE CTDD HILC EIHS ITEG OEAO SDDR YDLO RITR KLML EHAO
GTDH ARDP NEOH MGFM FEUH EEC D MRIP FEIM EHNL SST RTVD OHW?

This matrix allows us to duplicate the ciphertext that appears on the sculpture, and may explain the offset characters END|YAHR found on the sculpture as a clue for discovering the original matrix. More importantly, it shows that the question mark preceded the original plaintext message and may have been a continuation of K2.

ALTERNATE TRANSPOSITION TEST

A Transposition of the letter “M” from pos.169 to pos.337...

```
0 ?END YAHR OHNL SRHE OCPT EOIB IDYS HNAI ACHT NREY ULDS LLSL LNO
1 HSNO SMRW XNET PRNG ATIH NRAR PESL NNEL EBLP IIAC AEWM TWN
2 DITE ENRA HCTE NEUD RETN HAEO ETFO LSED TIWE NHA E IOYT EYQ
3 HEEN CTAY CREI FTBR SPAM HHEW ENAT AMAT EGYE ERLB TEEF OASF IOT
4 UETU AEOT OARM AEER TNRT IBSE DDNI AAHT TMST EWPI EROA GRI
5 EWFE BAEC TDDH ILCE IHSI TEGO EAOS DDRY DLOR ITRK LMLE HAG
6 TDHA RDPN EOHM GFMF EUHE ECDM RIPF EIME HNLS STTR TVDO HWM
```

```
0 ?END YAHR OHNL SRHE OCPT EOIB IDYS HNAI ACHT NREY ULDS LLSL LNO
3 HEEN CTAY CREI FTBR SPAM HHEW ENAT AMAT EGYE ERLB TEEF OASF IOT
6 TDHA RDPN EOHM GFMF EUHE ECDM RIPF EIME HNLS STTR TVDO HWM |
2 DITE ENRA HCTE NEUD RETN HAEO ETFO LSED TIWE NHA E IOYT EYQ |
5 EWFE BAEC TDDH ILCE IHSI TEGO EAOS DDRY DLOR ITRK LMLE HAG |
1 HSNO SMRW XNET PRNG ATIH NRAR PESL NNEL EBLP IIAC AEWM TWN |
4 UETU AEOT OARM AEER TNRT IBSE DDNI AAHT TMST EWPI EROA GRI |
| | | | | | | | | | | | |
```

Plaintext Sample:

ILG THE HOLE A LITTUE I INSENTED THE AANDLE AHD PEEREI
IN THE HET AIR ESOAPING FSOX THE COASBER CYUHED TH?

B Transposition of the “?” from pos.1 to pos.337:

```
0 ENDY AHRO HNLS RHEO CPTE OIBI DYSH NAIA CHTN REYU LDSL LSL NOH
1 SNOS MRWX MNET PRNG ATIH NRAR PESL NNEL EBLP IIAC AEWM TWN
2 DITE ENRA HCTE NEUD RETN HAEO ETFO LSED TIWE NHA E IOYT EYQ
3 HEEN CTAY CREI FTBR SPAM HHEW ENAT AMAT EGYE ERLB TEEF OASF IOT
4 UETU AEOT OARM AEER TNRT IBSE DDNI AAHT TMST EWPI EROA GRI
5 EWFE BAEC TDDH ILCE IHSI TEGO EAOS DDRY DLOR ITRK LMLE HAG
6 TDHA RDPN EOHM GFMF EUHE ECDM RIPF EIME HNLS STTR TVDO HW?
```

```
0 ENDY AHRO HNLS RHEO CPTE OIBI DYSH NAIA CHTN REYU LDSL LSL NOH
3 HEEN CTAY CREI FTBR SPAM HHEW ENAT AMAT EGYE ERLB TEEF OASF IOT
6 TDHA RDPN EOHM GFMF EUHE ECDM RIPF EIME HNLS STTR TVDO HW? |
2 DITE ENRA HCTE NEUD RETN HAEO ETFO LSED TIWE NHA E IOYT EYQ |
5 EWFE BAEC TDDH ILCE IHSI TEGO EAOS DDRY DLOR ITRK LMLE HAG |
1 SNOS MRWX MNET PRNG ATIH NRAR PESL NNEL EBLP IIAC AEWM TWN |
4 UETU AEOT OARM AEER TNRT IBSE DDNI AAHT TMST EWPI EROA GRI |
| | | | | | | | | | | | |
```

Plaintext Sample:

ING THE HOLE A LITTLE I INSERTED THE CANDLE AND PEERED
IN THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE

PLAINTEXT MESSAGES

K1 Plaintext:

Between subtle shading and the absence of light lies the nuance of **iqclusion**.

Revised K2 plaintext:

John: It was totally invisible.

Jane: How's that possible?

John: They used the earth's magnetic field.

xxxpausexxx

John: The information was gathered and transmitted 'underground' to an unknown location.

xxxpausexxx

Jane: Does Langley know about this?

John: They should; it's buried out there somewhere.

xxxpausexxx

Jane: Who knows the exact location?

John: Only WW. This was his last message...

xxxpausexxx

+38° 57' 6.5" North, -77° 8' 44" West

Jane: Layer two?

Revised K3 plaintext:

Slowly, **desparatly** slowly, the remains of passage debris that encumbered the lower part of the doorway **was** removed. With trembling hands, I made a tiny breach in the upper left-hand corner and, then, widening the hole a little, I inserted the candle and peered in. The hot air escaping from the chamber caused the flame to flicker but, presently, details of the room within emerged from the mist.

xxxbreakxxx

Can you see anything Q